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Reuse Selection Criteria for Progressive Cavity Pumps Based on the Mean Time between Failure and Reliability Function for Each Pump Type in the Cira Infantas Field

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Abstract

La Cira-Infantas (LCI) is Colombia's oldest oil field whose production started on April 18th, 1918. Since it is a mature field and is undergoing secondary recovery most of its wells have low flow rates (< 1000 BFPD). The wells require an artificial lifting system with low-cost installation and maintenance. For low flow and gassy wells in sandstone reservoirs, the Progressive Cavity Pump (PCP) system has been used since 2006.

It has been observed that the smallest percentage of PCP failures is due to the pump. When a pump fails, it is necessary to determine if a component should be reused. Currently, when any work is performed on a well, the pump is inspected with a borescope. When the results show that the pump efficiency is 50% or higher, it is considered reusable. In addition, corrosion, scale formation, wear and abrasion from reservoir fluids should be identified to determine if the pump can be reused.

A database was constructed of the history of the PCP in LCI. Each well had its run life, type of installed pump and the cause for any intervention. Thus, every pump model was classified, identifying all the interventions as pump failure or no-failure using failure codes. This data was used to obtain the Mean Time Between Failure (MTBF) calculation and its corresponding probability of survival, obtaining the initial criteria to reuse the pumps according to what was calculated and the experience and knowledge of the conditions in the field. This history, combined with the test bench efficiency analysis, assisted in determining if the pump was in good condition and a candidate for reinstallation.

Today, when a pump is pulled out and successfully passes testing, its information is compared against every pump in the database. This enables a prediction of how much longer the PCP will last. After the estimate of reliability is calculated, the decision is made whether to reuse it or to reject it.

Introduction

For PCP's, pump failures have never been a major issue considering that pump failures average 20% of the total failures since 2007. Nevertheless, in order to optimize costs, a selection criterion for reusing or discarding pumps was developed. The purpose of this study is to determine the reliability for each pump

model used in LCI in order to know in which case components can be reused and will likely last for more than 365 days or have to be discarded in order to avoid an early failure.

At the beginning the pumps were reused if they were found to be in good mechanical condition and a test bench showed good efficiency. This selection criterion was not enough and pumps failed in a short time if they already had a high run life. So, another selection criterion was developed to decide if used pumps could be rerun. A Mean Time Between Failure (MTBF) methodology was implemented for each pump model to determine if a pump could be reused according to the performance of the pump population in the field.

Field overview

La Cira-Infantas is an oilfield with black oil with an average API gravity of 22° producing from three main zones called A, B and C. The primary drive mechanism is solution gas drive. The gas was initially in a sub saturated condition but rapidly became saturated creating gas zones as the reservoir pressure was lowered. The C Zone, which has the higher oil reserves, is currently water flooded. Water flooding has introduced new operating problems such as corrosion, scale deposition and fines migration. Field data for pressure, temperature and volume (PVT) are shown in [Tables 1 and 2](#).

Table 1—Field PVT for La Cira

ZONE	RESERVOIR TEMPERATURE Ty (°F)	SATURATION PRESSURE (Ps)	OIL FORMATION VOLUME FACTOR AT Ps, Ty	Rs (CF/STB)	API	OIL VISCOSITY (cps)
A	92	526	1.04	70	18-24	30 cps at 100°F
B	98	650	1.047	113	20-27	16.5 cps at 98°F and 650 psi
C	105	1570	1.068	100	24	19 cps at 105°F and 1570 psi

Table 2—Field PVT for Infantas

ZONE	RESERVOIR TEMPERATURE Ty (°F)	SATURATION PRESSURE (Ps)	OIL FORMATION VOLUME FACTOR AT Ps, Ty	Rs (CF/STB)	API	OIL VISCOSITY (cps)
C	115	945	1.1	200	28	5.5 cps

PCP in LCI

LCI is a field located in the Middle Magdalena Valley in Colombia. The PCP Project started in 2006 with 13 wells with an average well production of 375 BFPD. The installation of this system started due to the incremental failures in sucker rod pumping systems (main artificial lift system in the field) because of high sand production that caused pumps to plug. Later, wells with production potential over 1000 BFPD could not be produced with beam pumps (BP). So, PCP's were installed as a good choice for handling sand at these flow rates.

The population of PCP's has been increasing year by year. The failure index, [Equation 1](#), (which is calculated as the amount of PCP failures divided by the average number of active PCP wells in one year) shows a significant increase in the first 4 years because of the learning curve associated with the new pumping system in the field as shown in [Figure 1](#).

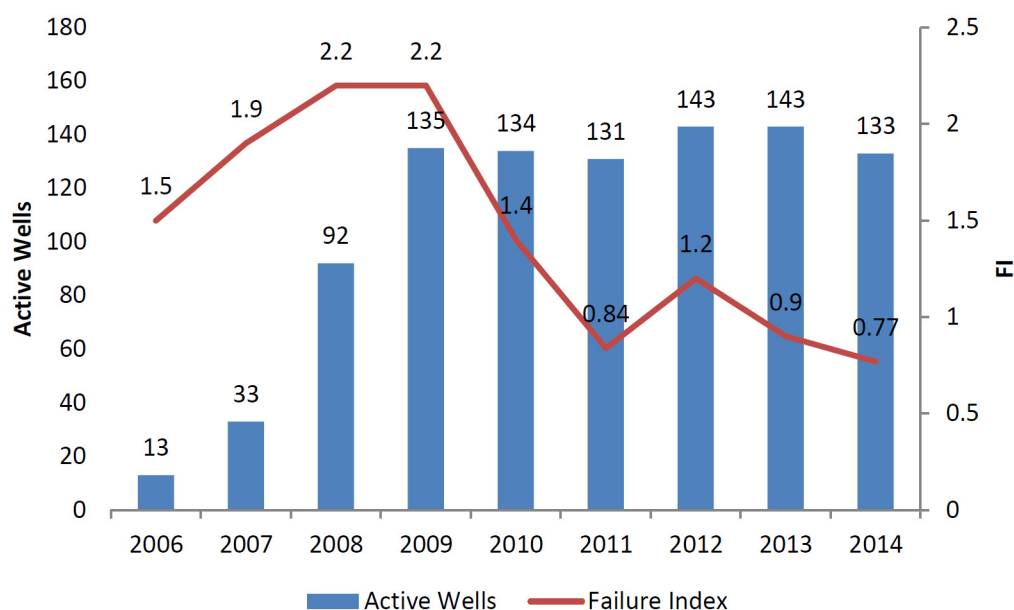


Figure 1—Failure Index for the PCP system in LCI Field

$$\text{Failure Index} = (\Sigma \text{Failures}) / (\text{Avg number of active wells}) \quad (1)$$

PCP failures in La Cira-Infantas Field are mainly due to rods and tubing failures. Pump failures account for a small percentage of the total failures. Consequently, the failure index has not been significantly affected by pump failures. The pumps have shown a rising run life trend since 2007, however this is not the performance expected for the main component of the system. In 2012, a downturn in performance for PCP's was caused by an increase of the failure index and poor pump run life (20% lower than in 2011).

For producing wells with PCP's, five different pump manufacturers were chosen from around the world to be used in LCI, but only three brands were analyzed according to the MTBF methodology because they, had the majority of the installations in the field. The root causes for these pump failures are classified in Figure 2.

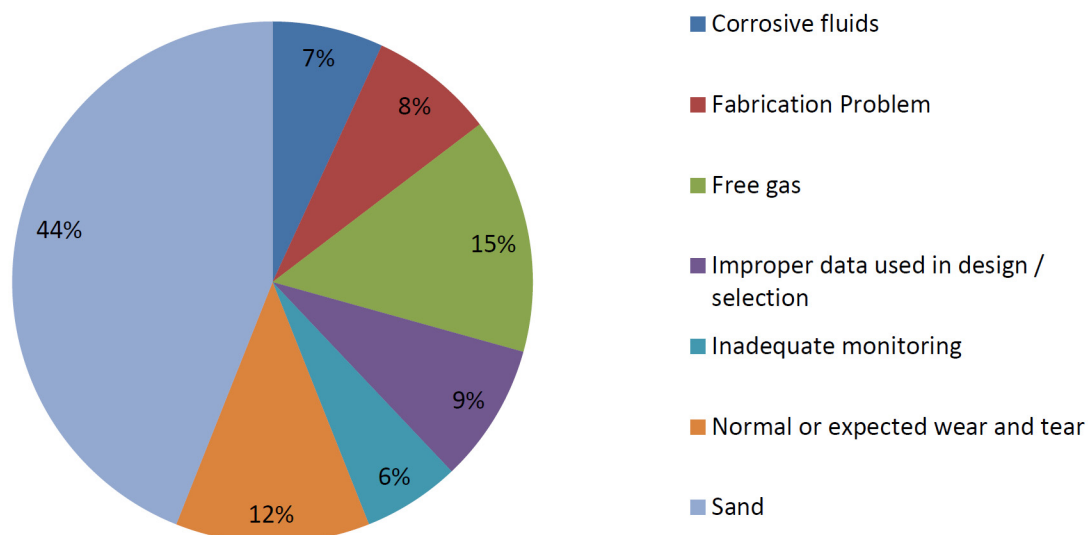


Figure 2—Root cause of PC pump failures

In most PCP rod or tubing failures, the pump can be checked in order to determine if can be reused to optimize costs. However, it has to be done carefully or it can result in a premature failure related to the pump.

Reuse of PCP's

Because La Cira-Infantas is being water flooded, the produced fluids contain high water cuts with low pH and high oxygen levels that cause corrosion. Also, scale formation is an issue because of the high concentration of calcium and magnesium detected in the injection water as well as the incompatibility of the formation and make-up fresh water that cause the formation of CaCO_3 and MgCO_3 scale. These particular fluid conditions make the reusing of pumps a concern. Serious work must be done in order to avoid failures from reusing equipment and a cost optimization is required.

When a PCP is pulled, the first step at the well site is to evaluate the stator condition (exterior and interior), rotor, non turn tool, swivel, gas separator and tag bar. Special attention is given to finding corrosion and/or wear. Then, the stator is fully inspected with a borescope, an optical device used to inspect an inaccessible space, to identify any damage that might be present in the elastomer. Conditions such as scratches, swelling, blisters and erosion are looked for. Based on this inspection, the field engineer takes a decision to:

1. Reuse the pump if it is in good condition.
2. Send it to the workshop for a bench test where the pump is tested on a bench at different speeds (RPM) for different discharge pressures. The flow rate, torque and efficiency are measured in order to determine the pump condition. When a pump performs with an efficiency between 50 and 80%, it is considered reusable. If the pump efficiency is below 50%, it is discarded.
3. Discard the pump. The pump is discarded if it is found to be deteriorated or has an efficiency lower than 50% based on the production test or test bench.

Prior to 2013, only these reuse criteria were used. The reuse of pumps using these criteria caused an increase in pump failures as a consequence of the reuse of medium to high run lives. As a result, pumps failed with in less than 365 days. In 2014, it became obvious that a new criterion was needed to statistically determine the survival probability of a pump according to historical run lives.

MTBF Methodology and survival probability for PC pumps

With the purpose of establishing a successful reuse criteria for PC pumps for each pump model in the LCI field, the mean time between failures (MTBF) methodology¹ was used. The MTBF is given in [Equation 2](#), which is a reliability measurement for a product (average time between failures). It is considered as the total operating time of a population of equipment divided by the total number of failures of repairable equipment. The calculation of the MTBF uses [equations 2, 3, 4 and 5](#), as follows:

$$MTBF = \sum_{i=1}^N R_{ti} * (t_i - t_{i-1}) \quad (2)$$

Failure (1):

$$R_{ti} = \frac{(DP+1) - N_i}{(DP+2) - N_i} * R_{ti-1} \quad (3)$$

Failure (0):

$$R_{ti} = R_{ti-1} \quad (4)$$

Where:

DP = Total number of installations.

N = Number of installations

t = Operating time

$$Ps_i = e^{-\left(\frac{t_i}{MTBF}\right)} \quad (5)$$

Where:

PS_i = Survival probability.

As mentioned at the beginning of this paper, when the project started the pump models for each manufacturer were not standardized. Because several pump models were found to have few installations, the analysis was focused on the pump models with more than 20 installations in the field. Around 15 pump models fit into that category.

The data gathering for the MTBF calculation require the consolidation of all the information for each pump model and the definition of the failure codes. If a pump failed, a Code 1 is assigned. Otherwise, for rod or tubing failures or no failure a code 0 is assigned. The MTBF variables are then calculated and a survival probability curve is graphed for each pump model (Figure 4). The MTBF is then calculated for each model and the survival percentage is determined. This information is then used to decide if the pump can be reused with low risk or if it needs to be discarded.

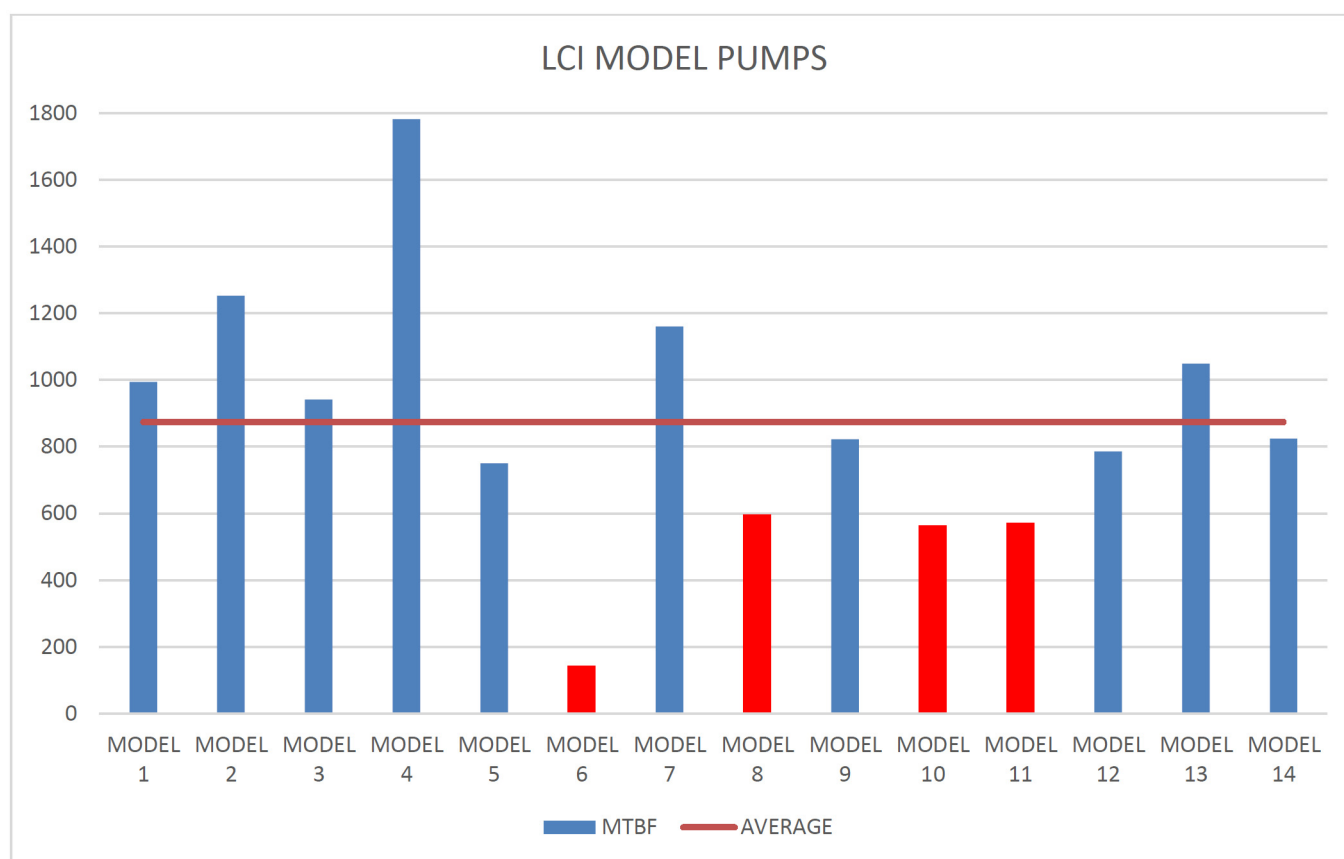


Figure 3—LCI Model Pumps

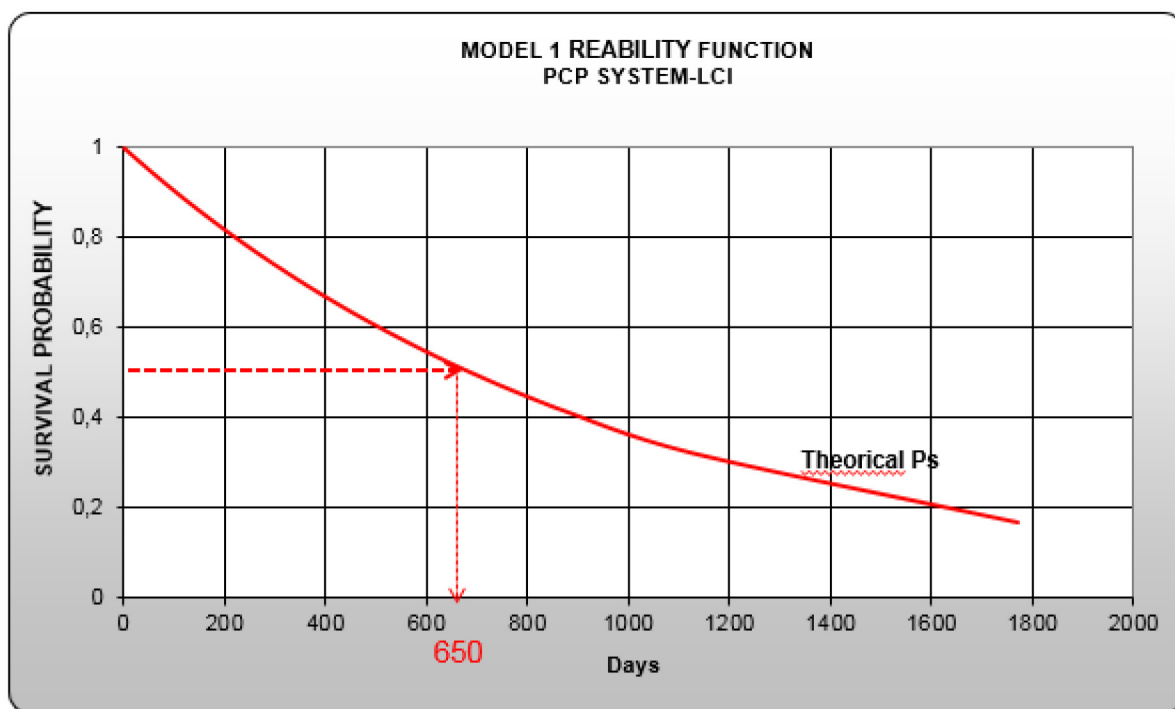


Figure 4—Reliability function

For the analyzed pumps, a MTBF average was calculated. It was found that the MTBF was 874 days for the pump models analyzed in LCI. Considering the good average MTBF obtained, 730 days was selected as the minimum MTBF required for a pump to be reused. If the MTBF for a pump model is less than 730 days, this pump shall never be reused. These pumps models appear in red in Figure 3.

As starting point in LCI, a MTBF requirement of at least 730 days was established with a minimum 365-day survival probability over 50%.

For example, for the pump Model 1 case, the MTBF calculated was 994 days and the 50% pump life survival probability was 650 days.

If a Model 1 pump is pulled with a run time of 200 days and it is known that the reliability for this pump is 650 days, we can expect that this pump can run at least another 450 days with a reliability percentage above 50%. This is higher than one year and the pump can be reused if found in good conditions. If this pump was pulled after running for 450 days, the risk would have been too high for reusing it and the pump would have been discarded.

Based on the previous calculation, the reuse criteria for PC pumps is defined as follows:

1. Pump model identification.
2. Visual inspection at the field location. If the pump is in good condition, use the borescope.
3. If the pump is in good condition as verified with the borescope, it is sent to the test bench. If it has an efficiency above 50%, knowing that the efficiency didn't decrease during operation, it is suitable for reuse.
4. If the MTBF of the pump model is higher than 730 days and if the probability of having more than 365 days of remaining life is greater than 50%, the pump can be reused. Otherwise, it should be discarded.
5. If the pump is visually in bad condition with the borescope, it should be discarded.

Conclusions

- I. LCI field has conditions that make PCP system installations suitable. However, special care must be taken when reusing pumps due to corrosion, scale formation, wear and abrasion.
- II. The failure index for PCP's in LCI field has been decreasing where improvements in system elements have been enforced and a better selection criterion for reusing components has been applied.
- III. It has been determined that bench testing and a field inspection with borescope are not enough to determine if a pump should be reused. In addition, this information must be combined with statistical data, and a MTBF calculation.
- IV. Using reliability curves for each pump model utilized in the field allows for a reliable reuse of pumps because, with the use of historical statistical data, there is better performance and equipment failure projections.

Acknowledgements

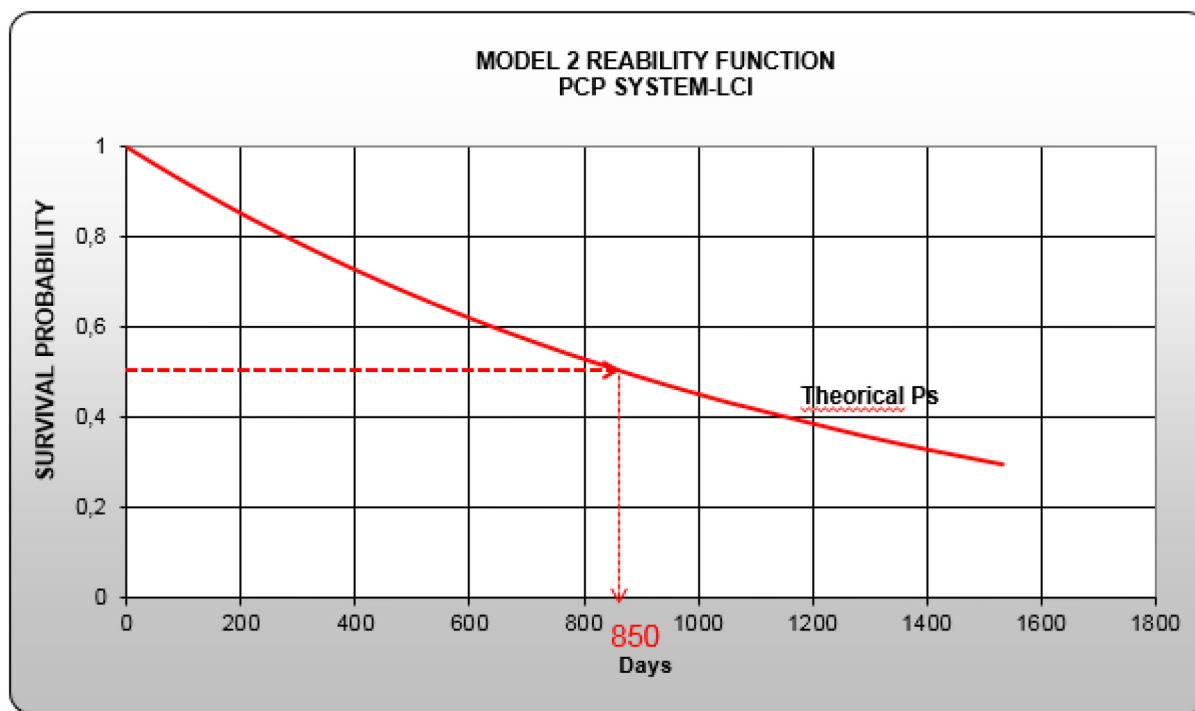
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References

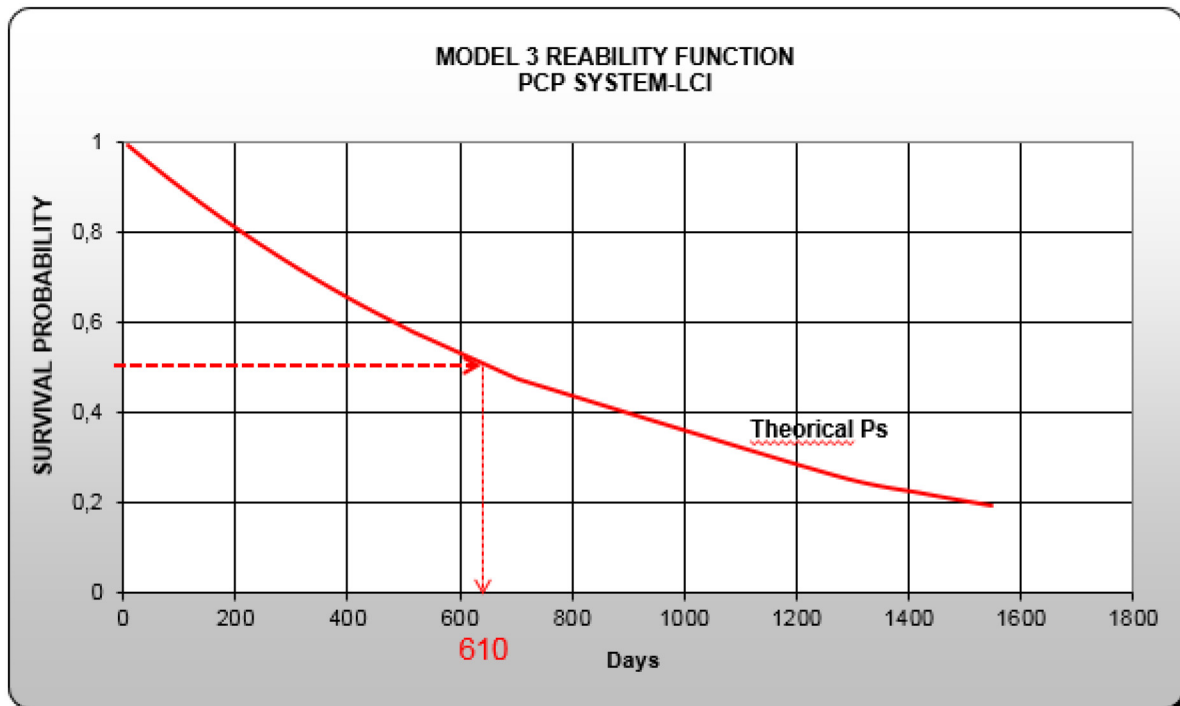
- W.J. Bailey, I.S. Weir, B. Couët and J. Hogan.: "Survival Analysis: The Statistically Rigorous Method for Analyzing Electrical Submersible Pump System Performance". Paper SPE 96722, paper presented at SPE Annual Technical Conference and Exhibition held in Dallas, Texas, U.S.A., 9 – 12 October 2005.
- S. Speaks, Vicor Reliability Engineering: "Reliability and MTBF Overview". pp: 2–10.
- Klein, J.P. and Moeschberger, M.L.: "Survival Analysis: Techniques for Censored and Truncated Data", New York: Springer, 1997.

Appendix A

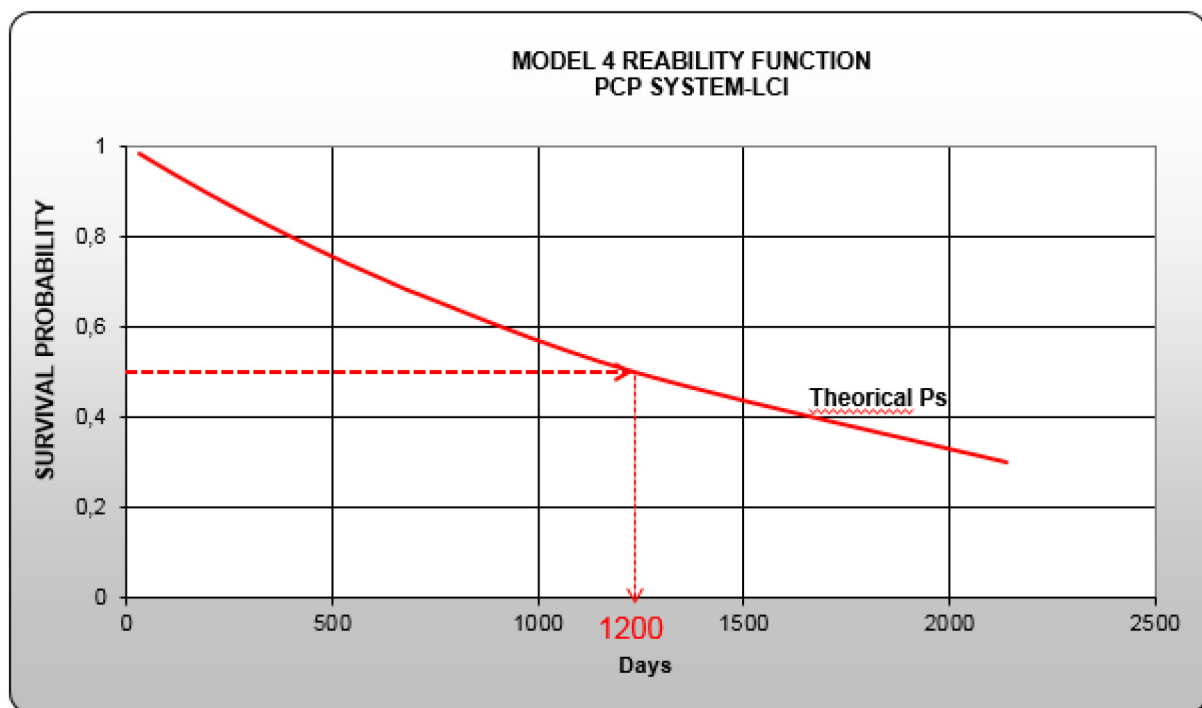
For the Model 2 case, the calculated MTBF was 1253 days and the 50% probability for this case was 850 days.



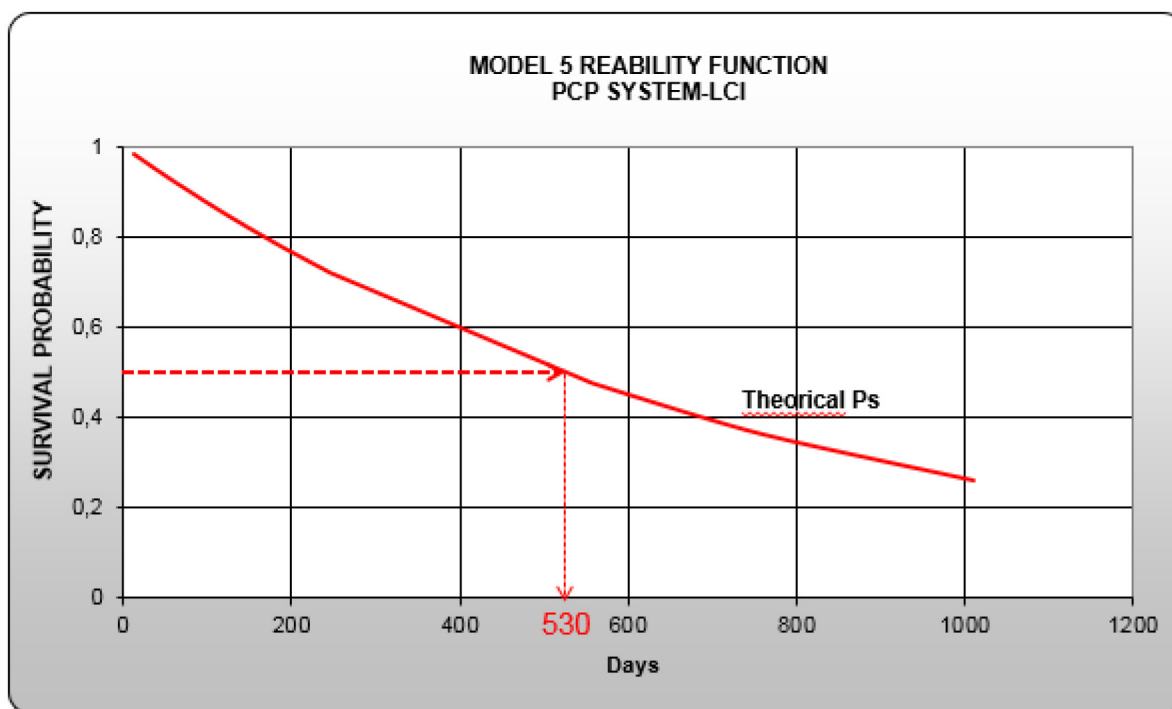
For the Model 3 case, the calculated MTBF was 941 days and the 50% probability for this case was 610 days.



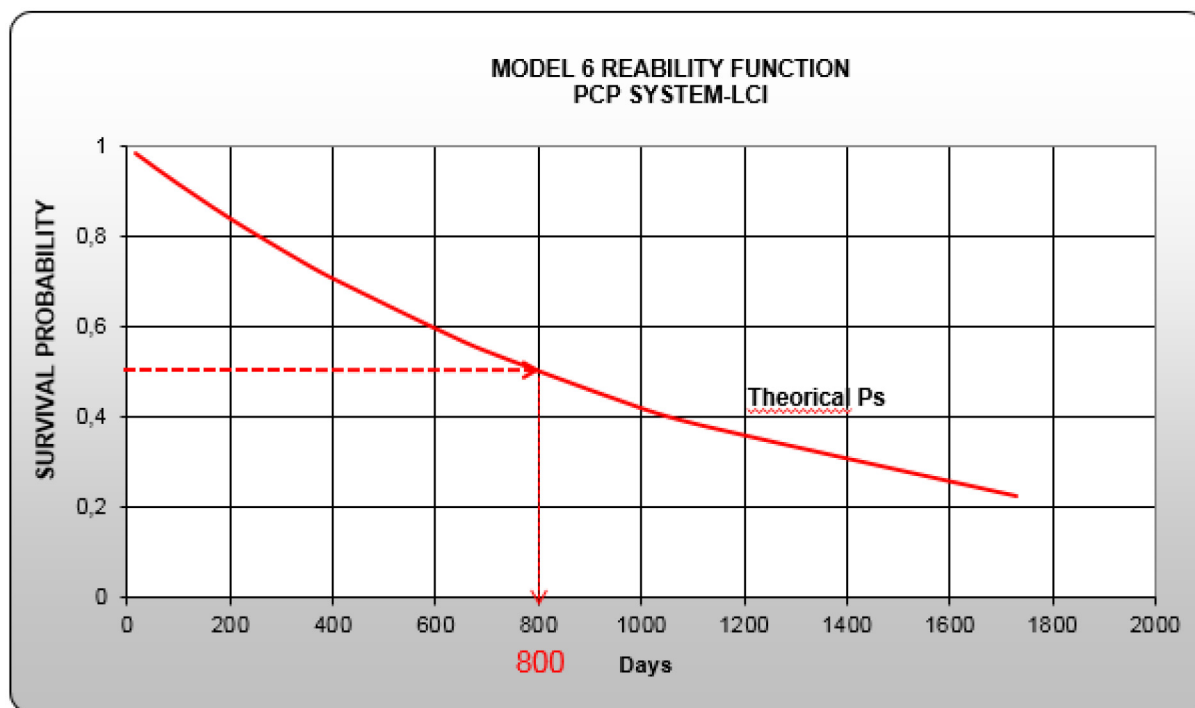
For the Model 4 case, the calculated MTBF was 1782 days and the 50% probability for this case was 1200 days.



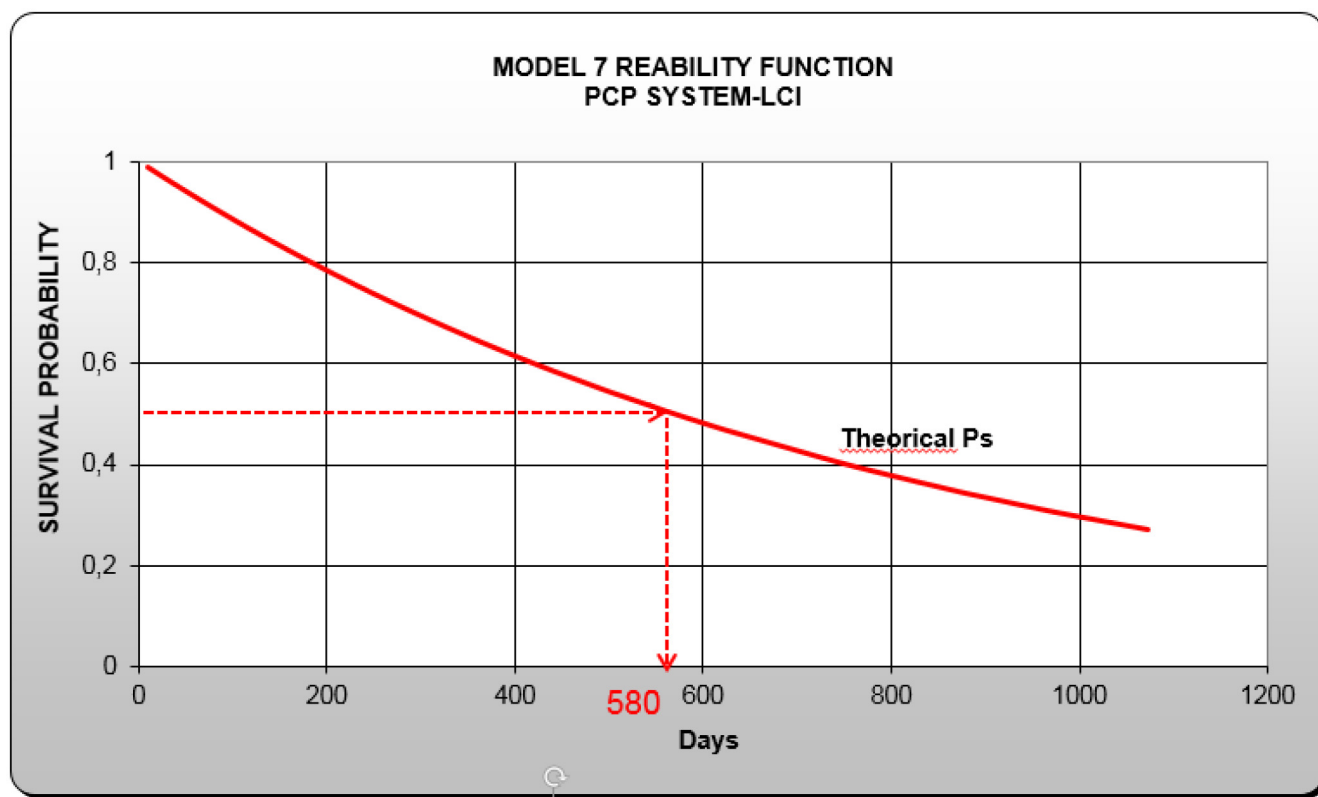
For the model 5 case, the calculated MTBF was 750 days and the 50% probability for this case was 530 days.



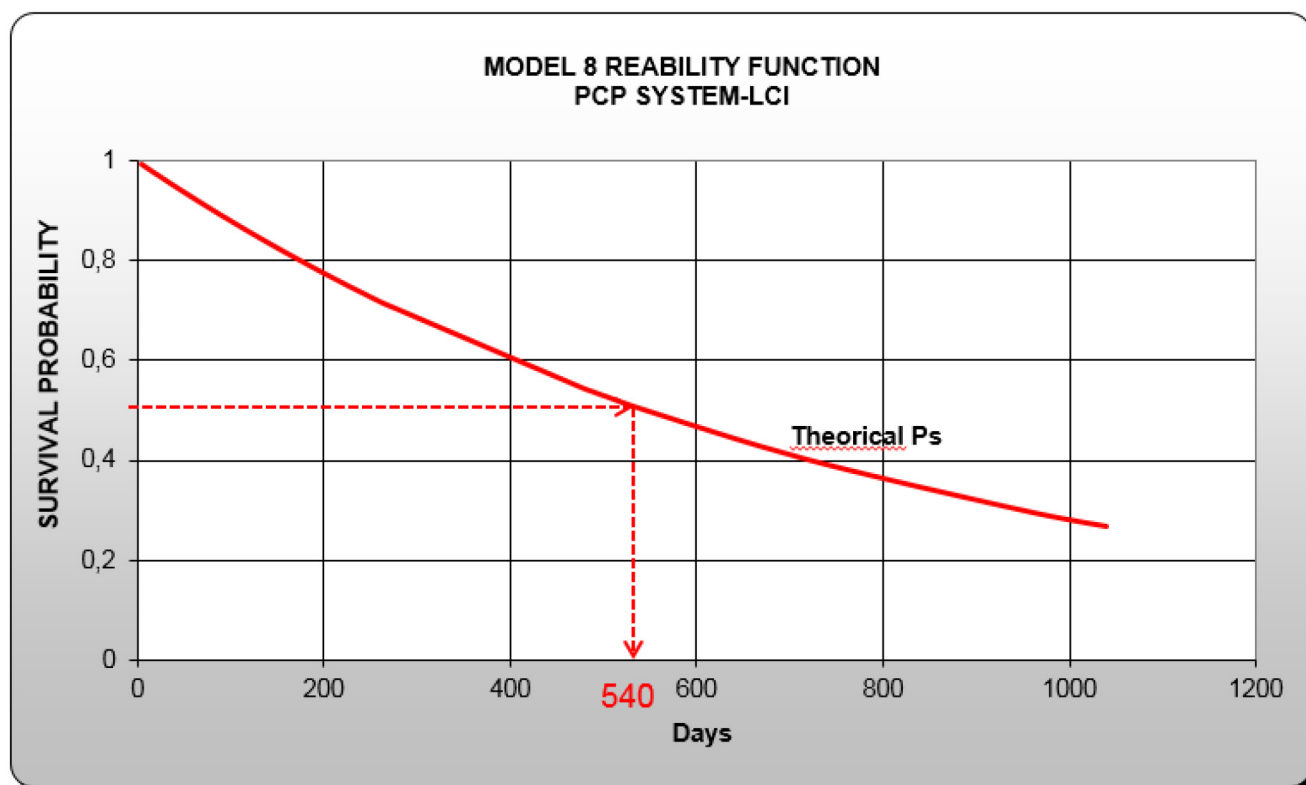
For the model 6 case, the calculated MTBF was 1160 days and the 50% probability for this case was 800 days.



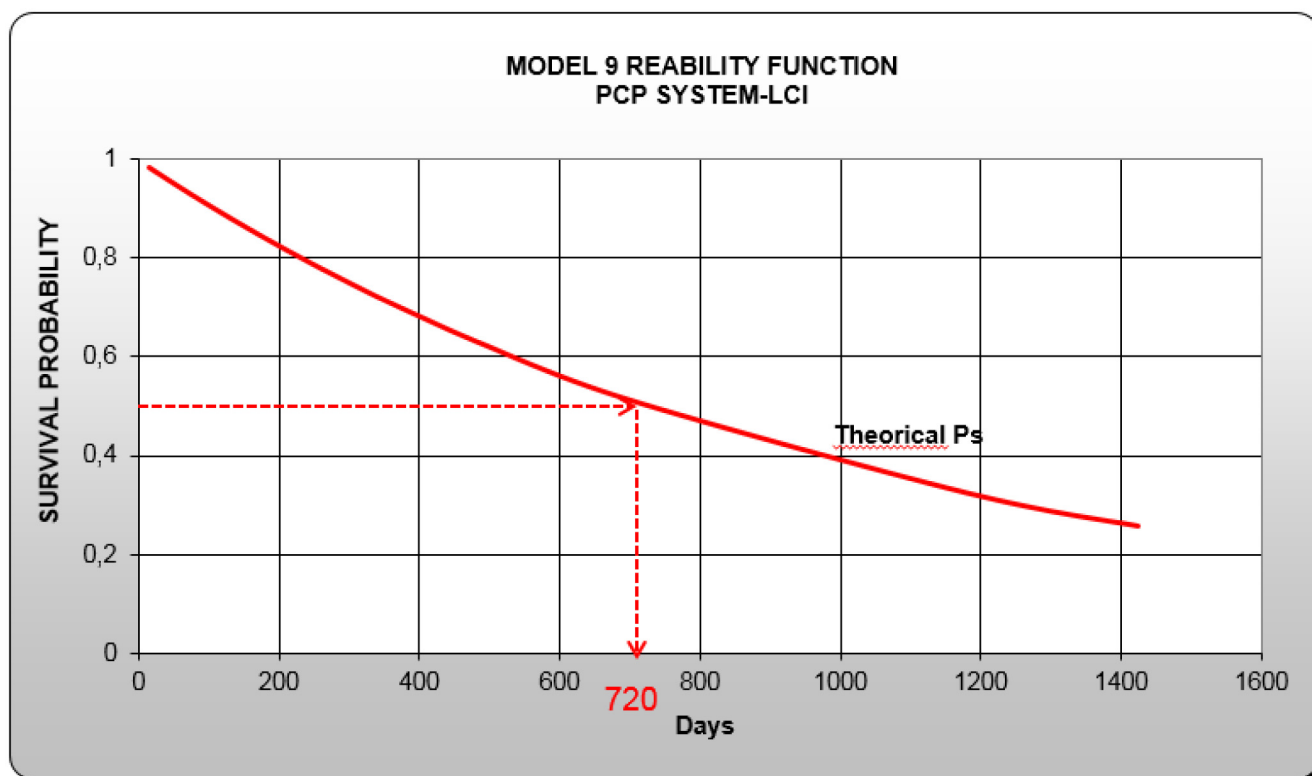
For the model 8 case, the calculated MTBF was 822 days and the 50% probability in for this case was 580 days.



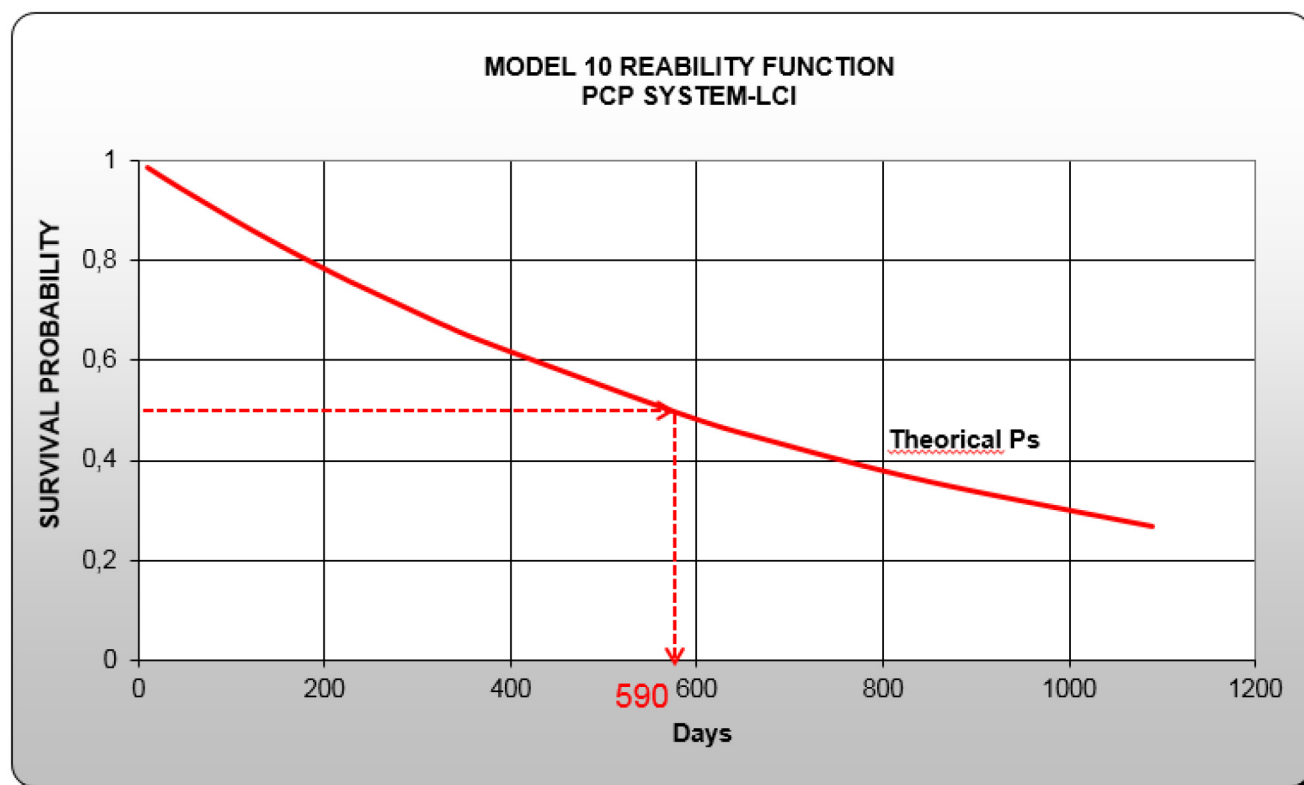
For the Model 8 case, the calculated MTBF was 786 days and the 50% probability for this case was 540 days.



For the Model 9 case, the calculated MTBF was 1049 days and the 50% probability for this case was 720 days.



For the Model 10 case, the calculated MTBF was 824 days and the 50% probability for this case was 590 days.



H7 Fichas del participante

Decisión de reutilización de una bomba

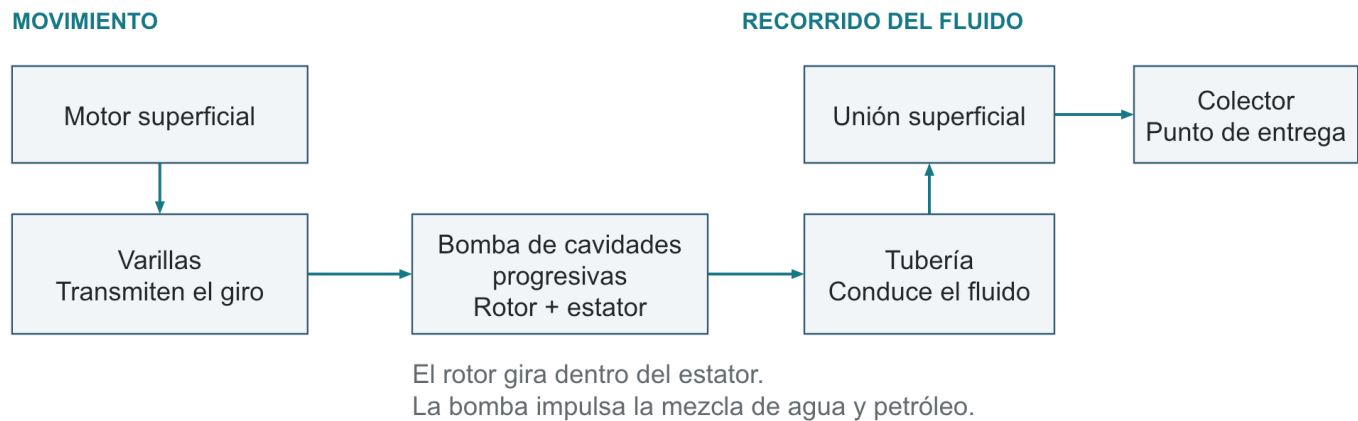
Trabajen en equipos de tres. Recomienden reutilizar, reemplazar o aplazar la decisión sobre una bomba recuperada. Justifiquen la respuesta y sus condiciones; después revisen el trabajo de otro equipo y ajusten su recomendación.

Equipo e integrantes

Caso real y adaptación del taller

El artículo de Labrador y colaboradores (2015) estudia la reutilización de bombas de cavidades progresivas en La Cira–Infantas. Combina inspecciones, pruebas de banco e historia de desempeño. Una bomba puede retirarse por una falla de varillas o tubería sin haber fallado ella misma.

Todos los datos de la bomba, fechas relativas, pruebas, requisitos, escenarios y tiempos de estas fichas son sintéticos. No corresponden a registros de La Cira–Infantas. Las hipótesis del equipo deben identificarse como tales.



PCP: bomba de cavidades progresivas. Rotor: pieza que gira dentro del estator fijo. Varillas: transmiten el giro. Colector: conducción superficial que recibe el fluido. Las flechas muestran movimiento y recorrido del fluido; no son un plano de construcción.

Servicio previsto	Dato sintético
Entrega	Al menos 100 m³/d de mezcla de agua y petróleo, a 3 bar en la entrada al colector.
Horizonte y condiciones	Campaña de 365 días con demanda continua. Fluido y energía disponibles; envolvente operativa definida por ingeniería.
Retorno	Entrega conforme, sin fugas y protecciones verificadas. Condiciones y resultados registrados.

Las páginas 1–3 contienen el expediente; las páginas 4–9, las fichas. Escriban en las celdas y líneas de Word, que se amplían al editar. No necesitan copiar la información de las diapositivas.

Fuente del caso de partida · SPE-173940-MS · 2015

Expediente de la bomba recuperada

Datos sintéticos · Diapositivas 6–8

Registro de la intervención

Momento	Hecho
Día 1 · 06:00	Con demanda activa, el caudal cae a cero. Se detiene el sistema.
Durante la intervención	Se documenta una rotura de varillas. Se retira la bomba recuperada.
Día 2 · 06:00	Se inicia la prueba con otra bomba y las varillas reparadas.
Día 2 · 08:00	Se verifica entrega conforme de 115 m ³ /d a 3 bar, sin fugas.
Día 2 · 12:30	Se cierra administrativamente la orden de trabajo.

La bomba que recuperó el servicio es otra unidad. La bomba recuperada quedó retirada y se evalúa para una instalación futura. La causa de la rotura de varillas no ha sido establecida.

Inspección y prueba de la bomba recuperada

Evidencia	Resultado
Inspección visual e interna	No se observan daños que excedan los criterios del informe.
Prueba de banco	115 ± 3 m ³ /d. Informe confirma carga y temperatura representativas del servicio.
Criterio de esta prueba	Mínimo 100 m ³ /d; sin fuga y funcionamiento estable en el ensayo.
Trazabilidad	200 días documentados en la última instalación; uso acumulado anterior desconocido.
Antes de reinstalar	Pendientes: aptitud para el destino y pruebas de integridad y protecciones.

Tres registros de operación

Unidad sintética	Tiempo observado	Final del registro
Bomba de historial 1	620 días desde nueva	Falla de la bomba confirmada.
Bomba de historial 2	300 días desde nueva	Sigue operando al cierre del seguimiento.
Bomba recuperada	200 días en esta instalación	Retiro por rotura de varillas; no se observó falla de la bomba. Uso previo desconocido.

La comparación de estos registros sirve para clasificar eventos y calidad de los datos. No se requiere calcular el tiempo medio entre fallas ni estimar la vida remanente. El uso previo desconocido de la recuperada no equivale a cero.

Protección prevista: detener el motor ante una sobrecarga, conforme al ajuste definido por ingeniería. Su prueba funcional sigue pendiente. Integridad de la conducción: capacidad de contener el fluido sin fugas. La verificación de ambas funciones debe quedar registrada.

Escenarios y regla de criticidad

Datos sintéticos · Diapositivas 10–12

El propósito es ordenar estudios de confiabilidad. Cada escenario conserva las condiciones que se describen. No se ha estimado su ocurrencia. La matriz no autoriza por sí sola la operación ni establece la urgencia actual.

Escenario sintético	Consecuencias declaradas	Restauración
Pérdida de caudal	Parada controlada, sin lesión ni liberación. Se pierde toda la entrega; no hay respaldo.	Cuadrilla y repuesto habituales.
Fuga contenida	Sin lesión; liberación retenida dentro de la instalación y limpieza. Entrega parcial.	Especialista adicional.
Fuga al exterior	Lesión reversible con atención médica; fluido en un canal de agua externo. Entrega parcial.	Especialista adicional.

La pérdida de caudal corresponde a la bomba bajo demanda activa. Las fugas corresponden a una unión superficial de la conducción. Son escenarios para analizar; no se afirma que ocurrieron durante la intervención documentada.

Rúbrica del curso

Criterio	Código 1	Código 2	Código 3
Personas	Sin lesión prevista.	Lesión reversible con atención médica.	Posible lesión permanente o muerte.
Ambiente	Sin liberación.	Liberación contenida dentro de la instalación y limpieza.	Afectación a un receptor fuera de la instalación.
Servicio	Entrega cubierta.	Entrega parcial.	Pérdida de la entrega requerida.
Restauración	Cuadrilla y repuesto habituales.	Especialista adicional.	Medios externos especiales.

Si no existe evidencia para un criterio, registren pendiente. No asignen un código por intuición.

Regla de decisión

Primero: un código 3 en personas o ambiente exige revisión específica prioritaria del escenario. No se compensa mediante otros criterios.

Después: para los demás escenarios, calculen $0,7 \times \text{servicio} + 0,3 \times \text{restauración}$. Mayor puntaje significa precedencia de estudio. Los pesos son preferencias declaradas del ejercicio, no probabilidades.

Responsabilidades dentro del equipo

Una persona revisa función y registros; otra, fallas y consecuencias; la tercera, la recomendación y su verificación. Compartan las conclusiones antes de decidir. Todos deben poder explicar la respuesta final.

Ficha 1 Función y lectura de los registros

Diapositivas 5–9 · Consulten el expediente de las páginas 1 y 2

1. Escriban la función completa y la frontera a la que se refiere

Momento que deben identificar	Respuesta del equipo
Inicio de la falla del servicio	
Recuperación funcional comprobada	
Cierre administrativo	
Indisponibilidad documentada y cálculo	

2. Clasifiquen cada registro. Pueden combinar etiquetas cuando describen aspectos diferentes.

Unidad	¿Se observó falla de la bomba?	Censura o historia incompleta y por qué
Historial 1 · 620 días		
Historial 2 · 300 días		
Recuperada · 200 días		

3. ¿Qué demuestra la prueba de banco y qué no permite concluir?

4. ¿Qué dato pendiente cambiaría su decisión y cómo lo obtendrían?

Recuerden distinguir el evento del sistema, el evento de la bomba y el final del seguimiento.

Ficha 2 Análisis de modos y efectos

Diapositiva 12 · Parte A · Completen con frases breves

FMEA o AMFE significa análisis de modos de falla y efectos. La observación está dada. Formulen los demás campos; identifiquen los mecanismos y las causas como hipótesis cuando no estén demostrados.

Campo	Bomba con entrega insuficiente	Unión superficial con fuga
Observación sintética	Se observan 85 m ³ /d con demanda activa y condiciones comparables.	Se observa fluido fuera de la unión.
Falla funcional		
Modo de falla		
Efecto local y en el servicio		
Mecanismo posible		
Causa posible y evidencia para contrastarla		
Acción y evidencia de cierre		

Para la primera columna, el expediente supone pérdida total de la entrega tras una parada controlada. Para la segunda, distinguan la ruta contenida y la ruta al exterior; sus consecuencias están en la página 3.

FMECA o AMFEC añade la criticidad a este análisis. El RPN, número de prioridad de riesgo, es severidad × ocurrencia × detección; no lo calculen aquí porque no se suministran las escalas ni la evidencia necesaria.

Ficha 2 Criticidad y barreras

Diapositivas 10–13 · Parte B

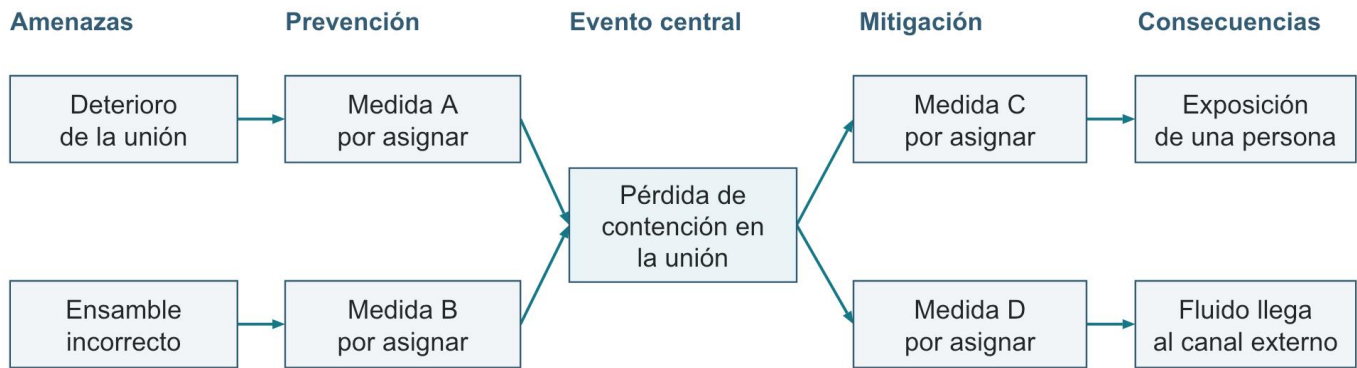
Justifiquen los códigos con los hechos de la página 3. La ocurrencia permanece no estimada.

Escenario	Personas	Ambiente	Servicio	Restauración	Tratamiento o puntaje
Pérdida de caudal					
Fuga contenida					
Fuga al exterior					

Hecho que activa la revisión específica y razón del orden restante

Bow Tie para la puesta en común

El corbatín representa amenazas, pérdida de control, consecuencias y barreras. Este complemento se revisa con el docente después del trabajo central.



Consecuencias posibles si las barreras no resultan eficaces. No se acredita independencia ni reducción numérica del riesgo.

Opciones: inspección con aceptación; ensamble verificado; detección y aislamiento oportunos; contención secundaria; cierre administrativo. Asignen A, B, C y D y excluyan la opción que no demuestra una barrera.

A	B	C	D

Elijan una barrera y señalen qué prueba o registro demostraría su eficacia

Ficha 3 Recomendación sobre la bomba

Diapositivas 7 y 15 · Parte A

Marquen o escriban una opción: reutilizar con condiciones / reemplazar / aplazar la decisión. No autoricen una reinstalación con condiciones esenciales pendientes. Expliquen cómo se resolverían.

Recomendación y dos evidencias del expediente que la sustentan

Condición o acción pendiente	Responsable propuesto	Resultado que permitiría avanzar

Cómo comprobarían el retorno al servicio y qué registrarían

Qué seguimiento proponen y qué modo de falla atiende

Qué dato o resultado les haría cambiar de recomendación

Cambiar la bomba no elimina por sí solo las condiciones de integridad de la conducción ni las pruebas de las protecciones. No se exige una optimización de costos.

Ficha 3 Respuesta a una señal

Diapositiva 14 · Parte B · Segunda situación del mismo sistema

Ya hay una bomba instalada y autorizada, independientemente de la decisión anterior. Durante el seguimiento se detecta una señal. Los datos siguientes son sintéticos y corresponden a condiciones comparables de presión, demanda y operación.

Horas respecto a D	-12	-8	-4	0
Caudal en m ³ /d	128	126	123	120

D es el momento de detección: el caudal llega al criterio de alerta de 120 m³/d. La función requiere al menos 100 m³/d. Una evaluación de ingeniería estima entre 8 y 12 horas desde D hasta la pérdida de función para este modo y condiciones. La ventana no se deduce de estos cuatro datos ni del tiempo medio entre fallas.

Actividad secuencial	Tiempo previsto
Confirmar	1 h
Decidir	0,5 h
Preparar	3 h
Ejecutar	2 h
Verificar	0,5 h

Margen adicional exigido por el ejercicio: 2 h. Todas las actividades empiezan después de D y se realizan en secuencia. No supongan actividades simultáneas sin justificarlas.

1. ¿Ya existe falla funcional al detectar la señal? Sustenten con el requisito

2. Tiempo total más margen y comparación con el límite inferior de la ventana

3. ¿Qué cambiarían si no cabe? Especifiquen la condición que deben demostrar

4. ¿Qué harían si esa condición no puede acreditarse?

D no necesariamente coincide con P, el momento en que el deterioro se vuelve detectable. La ventana restante desde D puede ser menor que el intervalo P–F.

Ficha 4 Revisión cruzada y respuesta final

Diapositivas 16–17 · Completa primero el equipo revisor

Equipo revisor y equipo revisado

Para cada criterio: 0 = ausente o contradictorio; 1 = parcial; 2 = claro y respaldado. La puntuación orienta la retroalimentación; el acuerdo con su propia decisión no es un criterio.

Criterio	0 a 2	Evidencia o mejora que señala el revisor
Función y lectura de registros		
Modos, efectos y causas posibles		
Criticidad justificada con los escenarios		
Decisión verificable y respuesta viable		

Una conclusión bien sustentada y el dato que la respalda

Una inferencia que requiere revisión y el motivo

Un cambio concreto recomendado

Respuesta final del equipo autor

Qué cambiamos después de la revisión o por qué mantenemos nuestra posición

Recomendación final acordada y nombres de los tres integrantes